MEMO

	Name	Company
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Subject:	Preliminary combined AOCS and TED time-series based on CM results	

Info:

The preliminary combined time series is based on the critical milestone TED delta assessment and the AOCS simulation WC_FPM_LONGRUN from the AOCS PDR part 2. The TED source data was ARC3 for the EOL case (rev2438_ARC3_EOL_90d_3600.mat) simulated for the critical milestone.

Note: the reaction wheel offloading assumptions used in the TED analysis are a fixed 3 day offloading cadence, however, the offloading in the WC_FPM_LONGRUN case are autonomous triggered by the RW speed limits. The RW speed evolution is mainly determined by the initial conditions and the evolution of the solar radiation pressure (and therefore the S/C attitude). The first FPM period was aligned to the end of the 14 hour tranquilisation period after the quarterly slew.

The NCAM33 w.r.t. FCAM2 was selected as the "worst case".

The first 14 hours of the TED data were removed to account for the tranquilisation time after the quarterly slewing.

The TED data were up-sampled to 8 Hz from 278 µHz (3600s sample period) using the Matlab function `griddedInterpolant` with the "Modified Akima cubic Hermite interpolation" interpolation method.

Due to the way re-initialisation of the FGS model worked when the AOCS WC_FP_LONGRUN was produced for AOCS PDR part 2, there was an artificially high change in the bias between each FPM segment. With the updated FGS model (v1.5) this change in bias after OLM is understood to not be present. A WC_FPM_LONGRUN with the updated FGS model is not yet available, therefore as a work around, the bias from each FPM segment was removed from the AOCS time series before combining with the TED data. If there is a change in the FGS bias that is expected after OLM that is not yet present in the FGS model v1.5, it could in principle be accounted for by adding a random bias of the appropriate size to each FPM segment. It is noted that the maximum bias change possible is constrained by PIRD-AOCS-022.

The AOCS WC_FPM_LONGRUN includes initial and trailing periods in GAM which were removed before combining with the TED data.

The combined data are contained in a file named `CM_LOS_Combined_v1.mat`, a matlab v7.3 format file based on hdf5 and should be readable with hdf5 tools.

The file contains three variables with dimensions as follows:

CM_LOS_Combined.mat (MAT-file)

Name	Value
LOSX	62729201x1 double
LOSY	62729201x1 double
LOSZ	62729201x1 double

Each LOS variable has a sample rate of 8Hz and is in radians.

The periods where reaction wheel offloading were simulated, including a tranquilisation time after the transition from OLM back to GAM are set to 'NaN'.

Note: Matlab uses column major storage.